

Voltage and current transformers

Lecture 16

Voltage and current transformers

- Transformers are probably the most critical elements in the production of electricity.
- It establishes a connection between two circuits, which appears to be a straightforward concept. But as the capacity of transformers expand, the idea becomes even more mind-blowing in its implications.
- The two most popular types of transformers are:
 - Voltage transformers.
 - Current transformers.

Voltage Transformers

- Voltage transformers are devices that are used to step down and step up (decrease/increase) the voltage of a system to a level that is considered safe.
- It operates by allowing energy meters to monitor the operation of electrical connections, which may have a higher voltage than usual to function correctly.
- Depending on the application, it can be used as a low voltage transformer or as a high voltage transformer.

Voltage transformer applications

The voltage transformer can be used as:

1. A useful instrument to measure the consumed energy.
2. A device to secure and prevent disturbances and faults from happening in the electrical systems.
3. Voltage signals are used crucially for synchronizing, event logging, and recording events.
4. A computer for measuring the impedance and power of a system in the process of voltage measurement along a direction.
5. A determiner for some fluxing security through the measurement processes.

Types of Voltage Transformers

1. Electromagnetic

This model of transformer is similar to the Oil-filled coiled wire because it uses a wire-wound core.

The primary voltage will be transformed entirely into the secondary voltage in this kind of converter, where the use of electromagnetic induction is accomplished.



Some applications to electromagnetic voltage transformers lie in their ability to:

- Employment in the field of electrical metering.
- Safeguard electrical instruments.
- Protect the feeder.
- Make generators protected from damage.
- Establish the electrification of industries

2. Capacitor

This kind of voltage transformer will be equipped with a capacitive potential divider, which will be linked between the main-line phase and the ground.

As a mixture of the capacitive voltage divider and a magnetic transformer, This converter can be as a hybrid device.

These transformers are referred to as intermediary converters with a relatively tiny ratio in the industry.

This type of transformer, which is composed of a stack of high-voltage capacitors, can be utilized for voltage metering in addition of being employed as a protection system.

The capacitors may reduce the voltage levels, and a transformer then steps down the lowered voltage to a supplementary voltage level.

3. Optical

Optical voltage transformers are a new type of power transformer that can be used for safety and metering applications instead of the usual electric-magnetic converters.

Many electric plant operators try to stay clear of them because of the high expenses and because the technology is relatively recent.

CHINT Voltage Transformer

The CHINT Voltage Transformer is a new and remarkable device for measuring voltages, power, and relay protection.

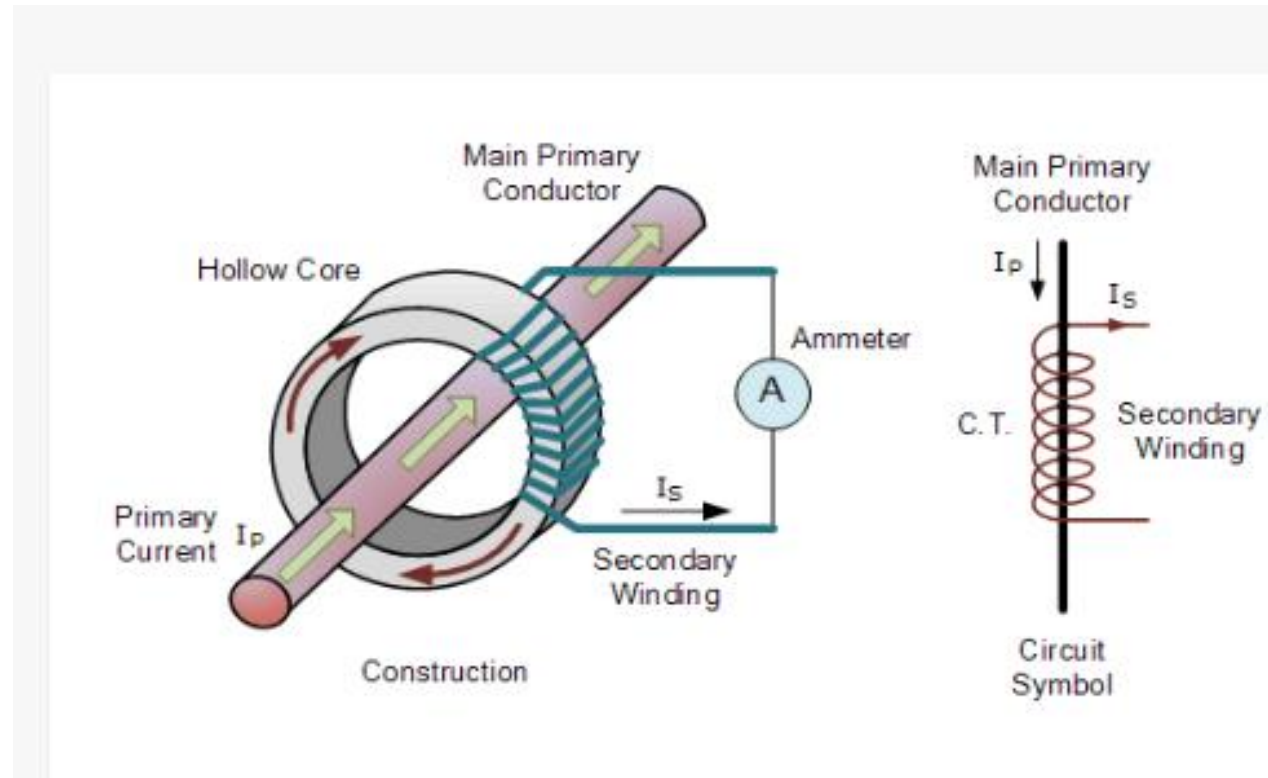
It has a reliable operation and insulation capability. It can also give a high measurement accuracy and reliable closure performance for industry applications. It's maintenance-free.



The Current Transformer

- The **Current Transformer** (**C.T.**), is a type of “instrument transformer” that is designed to produce an alternating current in its secondary winding, which is proportional to the current being measured in its primary.
- *Current transformers* reduce high voltage currents to a much lower value and provide a convenient way of safely monitoring the actual electrical current flowing in an AC transmission line using a standard ammeter.
- The principle of operation of a basic current transformer is slightly different from that of an ordinary voltage transformer.

The current transformer consists of only one or very few turns as its primary winding. This primary winding can be of either a single flat turn, a coil of heavy-duty wire wrapped around the core or just a conductor or bus bar placed through a central hole as shown in the following Figure:



Due to this type of arrangement, the current transformer is often referred too as a “**series transformer**” as the primary winding, which never has more than a very few turns, is in series with the current carrying conductor supplying a load.

The secondary winding may have a large number of coil turns wound on a laminated core of low-loss magnetic material.

This core has a large cross-sectional area so that the magnetic flux density created is low using much smaller cross-sectional area wire, depending upon how much the current must be stepped down as it tries to output a constant current, independent of the connected load.

The secondary winding will supply a current into either a short circuit, in the form of an ammeter, or into a resistive load until the voltage induced in the secondary is big enough to saturate the core or cause failure from excessive voltage breakdown.

Unlike a voltage transformer, the primary current of a current transformer is not dependent on the secondary load current but instead is controlled by an external load.

The secondary current is usually rated at a standard 1 Ampere or 5 Amperes for larger primary current ratings.

Types of current transformers

There are three basic types of current transformers:

1. **Wound** current transformers:

The transformer's primary winding is **physically connected in series with the conductor that carries the measured current flowing in the circuit.**

The magnitude of the secondary current is dependent on the turns ratio of the transformer.

2. **Toroidal** Current transformer:

These current transformers do not contain a primary winding.

Instead, the line that carries the current flowing in the network is threaded through a window or hole in the toroidal transformer.

Some current transformers have a “split core” which allows it to be opened, installed, and closed, without disconnecting the circuit to which they are attached.

3. Bar-type Current Transformer:

This type of current transformer uses the actual cable or bus-bar of the main circuit as the primary winding, which is equivalent to a single turn.

They are fully insulated from the high operating voltage of the system and are usually bolted to the current-carrying device.

- **Current transformers** can reduce or “step-down” current levels from thousands of amperes down to a standard output of a known ratio to either 5 Amps or 1 Amp for normal operation.
- Thus, small and accurate instruments and control devices can be used with CT’s because they are insulated away from any high-voltage power lines.

There are a variety of metering applications and uses for current transformers, such as with Wattmeter’s, power factor meters, watt-hour meters, protective relays, or as trip coils in magnetic circuit breakers, or MCB’s.

Most current transformers have the standard secondary rating of 5 amps with the primary and secondary currents being expressed as a ratio such as 100/5.

This means that the primary current is 20 times greater than the secondary current, so when 100 amps is flowing in the primary conductor, it will result in 5 amps flowing in the secondary winding.

A current transformer of 500/5, will produce 5 amps in the secondary for 500 amps in the primary conductor, 100 times greater.

By increasing the number of secondary windings, N_s , the secondary current can be much smaller than the current in the primary circuit being measured because as N_s increases, I_s goes down by a proportional amount.

The number of turns and the current in the primary and secondary windings are related by an inverse proportion.

A current transformer, like any other transformer, must satisfy the amp-turn equation as this turns ratio is equal to:

$$\text{T.R.} = n = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

from which :

$$\text{secondary current, } I_S = I_P \left(\frac{N_P}{N_S} \right)$$

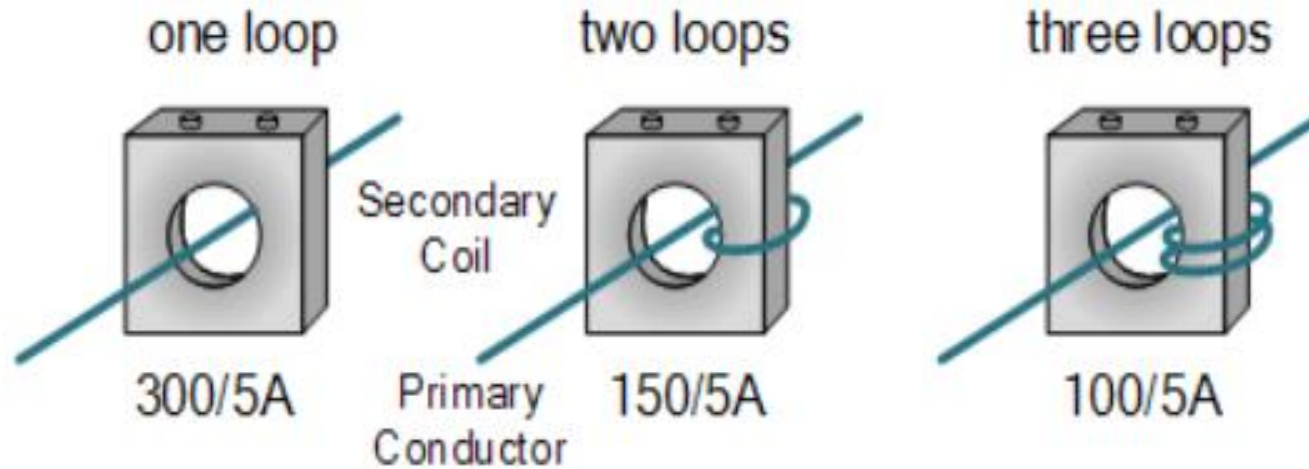
It should be noted that a current transformer rated as 100/5 is not the same as one rated as 20/1 or subdivisions of 100/5.

This is because the ratio of 100/5 expresses the “input/output current rating” and not the actual ratio of the primary to the secondary currents.

Also the number of turns and the current in the primary and secondary windings are related by an inverse proportion.

But large changes in a current transformers turns ratio can be achieved by modifying the primary turns through the CT's window where one primary turn is equal to one pass and more than one pass through the window results in the electrical ratio being modified.

For example, a current transformer with a relationship of, 300/5A can be converted to another of 150/5A or even 100/5A by passing the main primary conductor through its interior window two or three times as shown.



This allows a higher value current transformer to provide the maximum output current for the ammeter when used on smaller primary current lines.